

ME 333
MECHANICS OF MATERIALS

Lecture 4: Principal Strains & Strain Invariants

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Strain Measure

Q For a given Strain field is the displacement field unique?

- No. The reason can be found if we integrate the strain - displacement relations which give out constants of integration that are variable and which account for rigid body translations and rotations.

Strain Measure

Q What is the final direction of line element post deformation ?

- The engineering strain is defined as:

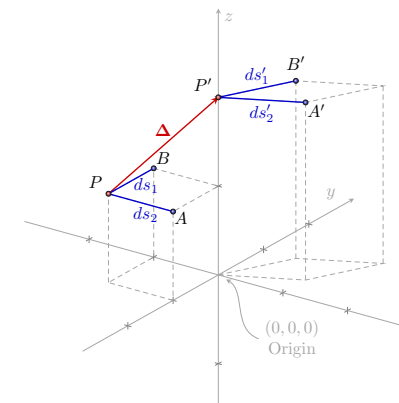
$$\epsilon_E = \left(\frac{ds' - ds}{ds} \right) = \frac{ds'}{ds} - 1 \implies \frac{ds'}{ds} = 1 + \epsilon_E$$

- Using relation between primed and unprimed quantities :

$$\begin{Bmatrix} \frac{dx'}{ds} \\ \frac{dy'}{ds} \\ \frac{dz'}{ds} \end{Bmatrix} = (1 + \epsilon_E) \begin{Bmatrix} l' \\ m' \\ n' \end{Bmatrix} = \begin{bmatrix} \left(1 + \frac{\partial u}{\partial x}\right) & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \left(1 + \frac{\partial v}{\partial y}\right) & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \left(1 + \frac{\partial w}{\partial z}\right) \end{bmatrix} \begin{Bmatrix} l \\ m \\ n \end{Bmatrix}$$

Strain Measure

Q Why is the y axis of Mohr's circle for strain $\gamma_{xy}/2$?



Strain Measure

- The angles between line elements in undeformed and deformed states:

$$\cos(\pi/2) = l_1 l_2 + m_1 m_2 + n_1 n_2 = 0; \quad \cos \theta' = l'_1 l'_2 + m'_1 m'_2 + n'_1 n'_2$$

- The shear strain measure can be written as:

$$\begin{aligned} \gamma_{xy} &= (1 + \epsilon_{Ex})(1 + \epsilon_{Ey}) \cos \theta' \\ &= 2l_1 l_2 \epsilon_{xx} + 2m_1 m_2 \epsilon_{yy} + 2n_1 n_2 \epsilon_{zz} + 2(l_1 m_2 + m_1 l_2) \epsilon_{xy} + 2(m_1 n_2 + n_1 m_2) \epsilon_{yz} \\ &\quad + 2(l_1 n_2 + n_1 l_2) \epsilon_{xz} \end{aligned}$$

- Special Case : If $l_1 = 1, m_1 = n_1 = 0$ and $l_2 = 1$ and $l_2 = n_2 = 0$, then:

$$\gamma_{xy} = 2\epsilon_{xy}$$

Note: Tensorial shear strain equals half of engineering shear strain.

Review of Basics of Tensors

Q What is a tensor ?

$$\begin{aligned} \mathbf{u} &= u \hat{\mathbf{i}} + v \hat{\mathbf{j}} + w \hat{\mathbf{k}} \\ \boldsymbol{\epsilon} &= \epsilon_{xx} \hat{\mathbf{i}} \otimes \hat{\mathbf{i}} + \epsilon_{yy} \hat{\mathbf{j}} \otimes \hat{\mathbf{j}} + \epsilon_{zz} \hat{\mathbf{k}} \otimes \hat{\mathbf{k}} + \epsilon_{xy} \hat{\mathbf{i}} \otimes \hat{\mathbf{j}} + \epsilon_{yz} \hat{\mathbf{j}} \otimes \hat{\mathbf{k}} + \epsilon_{zx} \hat{\mathbf{k}} \otimes \hat{\mathbf{i}} \\ &\quad + \epsilon_{yx} \hat{\mathbf{j}} \otimes \hat{\mathbf{i}} + \epsilon_{zy} \hat{\mathbf{k}} \otimes \hat{\mathbf{j}} + \epsilon_{xz} \hat{\mathbf{i}} \otimes \hat{\mathbf{k}} \end{aligned}$$

where:

$$\hat{\mathbf{i}} \otimes \hat{\mathbf{i}} = \begin{Bmatrix} 1 \\ 0 \\ 0 \end{Bmatrix} \begin{Bmatrix} 1 & 0 & 0 \end{Bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}; \quad \dots$$

[Example]

$$\boldsymbol{\epsilon} = \frac{\partial \mathbf{u}}{\partial \mathbf{x}} = \left(\frac{\partial}{\partial x} \hat{\mathbf{i}} + \frac{\partial}{\partial y} \hat{\mathbf{j}} + \frac{\partial}{\partial z} \hat{\mathbf{k}} \right) (u \hat{\mathbf{i}} + v \hat{\mathbf{j}} + w \hat{\mathbf{k}})$$

Review of Basics of Tensors

Coordinate Transformations

- Consider a vector $\mathbf{v}$:

$$\mathbf{v} = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} \begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \\ \mathbf{e}_3 \end{bmatrix} = \begin{bmatrix} v'_1 & v'_2 & v'_3 \end{bmatrix} \begin{bmatrix} \mathbf{e}'_1 \\ \mathbf{e}'_2 \\ \mathbf{e}'_3 \end{bmatrix}$$

- Taking dot product with respect to the primed unit vectors:

$$\mathbf{v}' = \begin{bmatrix} \mathbf{e}'_1 \cdot \mathbf{e}_1 & \mathbf{e}'_1 \cdot \mathbf{e}_2 & \mathbf{e}'_1 \cdot \mathbf{e}_3 \\ \mathbf{e}'_2 \cdot \mathbf{e}_1 & \mathbf{e}'_2 \cdot \mathbf{e}_2 & \mathbf{e}'_2 \cdot \mathbf{e}_3 \\ \mathbf{e}'_3 \cdot \mathbf{e}_1 & \mathbf{e}'_3 \cdot \mathbf{e}_2 & \mathbf{e}'_3 \cdot \mathbf{e}_3 \end{bmatrix} \mathbf{v} = \mathbf{Q}^T \mathbf{v}$$

Note: $\mathbf{Q}$ is transformation/rotation matrix and is not a *tensor*.

- The transformation matrix is orthogonal ($\mathbf{Q}^{-1} = \mathbf{Q}^T$) and therefore:

$$\mathbf{v} = \mathbf{Q} \mathbf{v}'$$

- For second order tensor therefore: $\mathbf{A}' = \mathbf{Q}^T \mathbf{A} \mathbf{Q}$

Review of Basics of Tensors

Tensor Invariants :

- Let us consider a symmetric second order tensor (3^2 elements) :

$$\mathbf{A} = \begin{bmatrix} a & e & f \\ e & b & g \\ f & g & c \end{bmatrix}$$

- Then its eigenvalues can be determined as:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} a - \lambda & e & f \\ e & b - \lambda & g \\ f & g & c - \lambda \end{vmatrix} = 0$$

- The eigen values represent the principal components. **WHY ?**

Review of Basics of Tensors

Tensor Invariants : Contd ...

- Its characteristic equation is:

$$\lambda^3 - I_1\lambda^2 + I_2\lambda - I_3 = 0$$

- The coefficients of the characteristic equation are termed as “Invariants” and are defined as:

$$I_1 = (a + b + c)$$

$$I_2 = (ab + ac + bc - e^2 - g^2 - f^2)$$

$$I_3 = (abc - ag^2 - e^2c - bf^2 + 2efg)$$

Review of Basics of Tensors

[Example 1] Determine the Invariants.

$$A = \begin{bmatrix} 5 & 2 & 4 \\ 2 & 3 & 1 \\ 4 & 1 & 6 \end{bmatrix}$$

- The invariants are:

$$I_1 = \text{tr}(A) = 5 + 3 + 6 = 14$$

$$I_2 = (15 + 30 + 18 - 4 - 1 - 16) = 42$$

$$I_3 = \det(A) = 29$$

Review of Basics of Tensors

[Example 1] Contd ...

Upon coordinate transformation (in plane rotation by 90°):

$$A' = Q^T A Q = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 2 & 4 \\ 2 & 3 & 1 \\ 4 & 1 & 6 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 5 & -4 \\ 1 & -4 & 6 \end{bmatrix}$$

- The invariants are:

$$I_1 = \text{tr}(A') = 3 + 5 + 6 = 14$$

$$I_2 = (15 + 18 + 30 - 4 - 16 - 1) = 42$$

$$I_3 = \det(A') = 29$$

Note: Invariants are quantities of a tensor that as the name suggests - *do not change* to coordinate transformations.

Review of Basics of Tensors

Q What is the invariant in the case of a first order tensor (vector)?

Review of Basics of Tensors

[Example 2] Invariant for a vector quantity

Let $v = \{2, 3, 4\}$ be any arbitrary vector. Its length (magnitude) is:

$$|v| = \sqrt{4 + 9 + 16} = \sqrt{29}$$

Let it transform through a rotation about the z-axis by 45° counterclockwise, then:

$$v' = Q^T v = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} 2 \\ 3 \\ 4 \end{Bmatrix} = \begin{Bmatrix} \frac{5}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 4 \end{Bmatrix}$$

Length $v' = \sqrt{29}$.

Review of Basics of Tensors

[Example 3] Coordinate transformations in 2D (Mohr's Circle)

- Let a tensor in planar case be defined as:

$$S = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}$$

- Then for any arbitrary rotation θ about the z-axis:

$$\begin{bmatrix} S'_{11} & S'_{12} \\ S'_{21} & S'_{22} \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

- The relations can be expanded and written as:

$$S'_{11} = S_{11} \cos^2 \theta + S_{22} \sin^2 \theta + S_{12} \sin 2\theta$$

$$S'_{22} = S_{11} \sin^2 \theta + S_{22} \cos^2 \theta + S_{12} \sin 2\theta$$

$$S'_{12} = (S_{22} - S_{11}) \sin \theta \cos \theta + S_{12} \cos 2\theta$$

Reading Assignment

Chapter 2: Sections 2.7 - 2.9